

Business Calculus (MATH 2350)

Summer 2018

Monday, 06/11/2018

Quiz 5: Section 11.6,11.7

Name (Print): _____

Score: _____ / 85

Section 11.6

1.

10 points

 If $x = x(y)$, $t = t(y)$ (that is, if both x and t represent functions of y), and $2\sqrt{x} = 11 + xt$, find $\frac{dt}{dy}$.

2. 10 points If $x = x(t)$, $y = y(t)$ (that is, if both x and y represent functions of t), and $3x^2 - 2y^3 = 11$, find $\frac{dx}{dt}$ when $x = 3$, $y = 2$, and $\frac{dy}{dt} = 6$.

3. 20 points A manufacturing company estimates that weekly sales s and weekly marketing costs m (both in dollars) are related by

$$s = 45,000 - 10,000e^{-.005m}$$

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The current weekly marketing costs are \$200, and these are increasing at a rate of \$7 per week. Find the current rate of change of weekly sales.

Section 11.7

1.

5 points

 Find the relative rate of change of $f(x) = 3 + 3e^{2x}$.
2.

5 points

 Find the relative rate of change of $f(x) = 6x^3 - \ln(x)$ at $x = 3$.

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3. 5 points Find the percentage rate of change of $f(x) = x^4 - 5$ at $x = 4$.
4. 10 points Given the price-demand equation $x = f(p) = 2000 - 3p^2$, find $E(p)$, the price-elasticity of demand. At $p = 15$, is demand elastic or inelastic?

5. Given the price-demand equation

$$3p^2 + 5x = 7500$$

(a) 5 points Express the demand x as a function of the price p .

(b) 5 points Find the elasticity of demand, $E(p) = \frac{-pf'(p)}{f(p)}$.

6. 10 points Given the price-demand equation $x = f(p) = \sqrt{200 - p^2}$, find the value(s) of the price p for which demand has unit elasticity.

7. 10 points (bonus) Given the price-demand equation $x = f(p) = 10e^{-0.1p}$, what price p maximizes the revenue?